

Multiple Choice (10 marks)

1. Find the Laplace transform of $f(t) = t^n$, where n is a positive integer,
 - (a) $L\{t^n\} = [n^2 / (s^{(n+2)})]$, for $s > 0$
 - (b) $L\{t^n\} = [n! / (s^{(n-1)})]$, for $s > n$
 - (c) $L\{t^n\} = [n! / (s^{(n+1)+1})]$, for $s > 0$
 - (d) $L\{t^n\} = [(n-1)! / s^{(n-1)}]$, for $s > 0$
 - (e) $L\{t^n\} = [(n+1)! / s^n]$, for $s > 0$
2. Determine the available energy that can be associated with a 10 m^3 perfect vacuum and its surroundings at a pressure of 0.100 MPa and at a temperature of 300 K.
 - (a) 750 kJ
 - (b) 1,500 kJ
 - (c) 250 kJ
 - (d) 2,000 kJ
 - (e) 1,250 kJ
3. Which lamp has the best Colour Rendering Index (CRI)?
 - (a) Xenon
 - (b) High pressure sodium vapour
 - (c) Halogen
 - (d) Low pressure sodium vapour
 - (e) Compact fluorescent
4. The rating of a certain machine from the name-plate is 110 volts, 38.5 amperes, 5 h.p. Find the input, output and efficiency at full load.
 - (a) Input: 4235 watts, Output: 3600 watts, Efficiency: 85 percent
 - (b) Input: 4500 watts, Output: 4000 watts, Efficiency: 88.9 percent
 - (c) Input: 4350 watts, Output: 3730 watts, Efficiency: 85.7 percent
 - (d) Input: 4100 watts, Output: 3450 watts, Efficiency: 84.1 percent
 - (e) Input: 4200 watts, Output: 3900 watts, Efficiency: 86 percent
5. The smallest change in measured variable to which instrument will respond is
 - (a) calibration.
 - (b) sensitivity.

- (c) resolution.
- (d) repeatability.
- (e) linearity.

True / False (10 marks)

Answer True or False

6. True or False: The solution for the equation $\dot{X} = AX + bu$ with the specified parameters results in $X(t) = 2$ for all t when $u(t)$ is the impulse function.
7. True or False: The speed of data transmission in a 4 G network is generally between 10 Gbps and 100 Gbps.
8. True or False: Delta winding is a necessary configuration for all DC machines, providing optimal performance under varying load conditions.
9. True or False: A 60-cycle alternator with a speed of 120 rpm must have 12 poles.
10. True or False: The average energy density of a plane wave with a Poynting vector of 5 watts/meter² is (1/4) erg/meter³.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the operational principles of an SR latch constructed from NOR gates and analyze the implications of the condition $S=1$ and $R=1$ in this context.
12. Analyze the probability of obtaining 2 heads and 1 tail when a penny is tossed three times. Discuss the underlying principles of probability that support your calculations.

End of Paper